
MINTS: Testing Strict Motif-Detector Claims in Nucleotide Transformers

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Abstract

1 Genomic foundation models often make biological labels linearly accessible, but
2 decodability alone does not identify an internal mechanism. We study whether
3 attention heads in nucleotide transformers can be justified as biological motif
4 detectors under a strict mechanistic evidence standard. MINTS applies tokenization-
5 aware QK/OV circuit analysis, residual-stream probing with controls, matched
6 attention enrichment, activation patching, and sparse feature search to DNABERT-2
7 and a tested Nucleotide Transformer backend. Before making mechanistic claims,
8 we report raw-sequence task baselines: GC-only AUROC ranges from 0.6361 to
9 0.9088, and 3–6-mer TF-IDF AUROC ranges from 0.7956 to 0.9406. DNABERT-2
10 layer-11 residual readouts achieve AUROC values of 0.9137, 0.9383, 0.8954, and
11 0.8847 on promoter and splice-site tasks, showing strong label decodability rather
12 than a circuit-level explanation. Controls sharpen this interpretation: random-
13 label and position-only probes are near chance; splice-site probes remain about
14 0.25 AUROC above GC-only controls; promoter probes are close to strong GC/k-
15 mer baselines and should be interpreted cautiously. The primary strict test is
16 negative: across 51,249 GM12878 CTCF sequences, no DNABERT-2 head meets
17 the registered QK-to-motif criterion or matched attention-enrichment criterion.
18 Batch activation patching finds promoter-TATA over-restoration and weaker splice-
19 donor restoration, but these task-specific causal signals do not establish a CTCF
20 motif-detector circuit. The results provide a calibrated evidence standard for
21 nucleotide-transformer interpretability and show where representation-level success
22 falls short of a strict motif-detector claim.

1 Introduction

24 Transformer models for nucleotide sequences now perform well on regulatory and splice-site pre-
25 diction tasks [8, 4, 7]. However, much of the usual evidence for biological understanding remains
26 predictive or post hoc: attribution maps, motif visualizations, or probes can identify useful correla-
27 tions, but they do not by themselves explain how internal computations produce predictions [1, 12].
28 This distinction is central for mechanistic interpretability, where a claim about a component should
29 survive direct tests of what it reads, where it attends, what it writes, and whether intervention on it
30 changes the relevant behavior [5, 10, 6].

31 Nucleotide transformers are a useful setting for this question. Known motifs give natural ground-truth
32 hypotheses: a motif can specify where a component would need to attend, what feature it would
33 need to read, and which perturbation should matter. They also expose a common interpretability risk.
34 Attention maps, motif visualizations, and high probe scores can suggest a biological mechanism even
35 when the model is using GC content, genomic-coordinate artifacts, distributed features, or another
36 correlated signal. MINTS is designed to make the required evidence explicit.

This paper contributes a calibrated evidence standard and a completed case study. MINTS separates three claims that are often conflated: residual decodability, task-level causal restoration under activation patching, and strict circuit-level motif detection. The primary strict test is CTCF: under the registered QK-alignment and matched-enrichment criteria, neither tested backend yields a CTCF attention head that should be called a strict motif detector. The promoter and splice analyses play a supporting role by showing that task labels can be decodable and that interventions can produce task-specific causal signals without establishing the stricter CTCF circuit claim.

2 Related Work and Positioning

Mechanistic interpretability of transformers has developed standard tools for decomposing attention heads, locating causal pathways, and tracing computational graphs [5, 10, 2]. QK/OV decomposition is especially relevant here because it separates where a head attends from what it writes. Activation patching then gives an intervention-based test of whether the component matters for a behavior [6]. MINTS applies these established tools to genomic transformers with explicit motif-level criteria, using ground-truth motifs to test both positive circuit evidence and possible overinterpretation.

Genomic deep-learning interpretability has a different tradition. Models such as DeepBind, SpliceAI, DNABERT, GPN, and Nucleotide Transformer show that neural sequence models learn useful biological regularities [1, 7, 8, 3, 4]. Many analyses in this area focus on motif visualization, attribution, saliency, or downstream probing. These are valuable, but they usually do not establish an internal algorithm. A saliency map may highlight a motif, and a probe may decode a label, while the model’s actual computation may depend on a distributed feature, a correlated composition signal, or a different component entirely.

MINTS is positioned between these literatures. It does not claim to solve genomic interpretability in general, and it does not treat a high downstream or probe score as mechanistic evidence. Instead, it asks a narrower question: when, if ever, is it justified to call a specific attention head a biological motif detector? This choice deliberately favors falsifiable criteria over narrative completeness. Negative results are therefore useful evidence, because they identify where an appealing mechanistic story is not supported by the implemented tests.

3 Methods and Evidence Criteria

Transformer circuits. Let $\tau(x) = (t_1, \dots, t_T)$ tokenize a nucleotide sequence into vocabulary items. The initial hidden state is

$$H_{j,:}^{(0)} = E_{\text{tok}}[t_j, :] + P_{j,:} \quad (1)$$

For head h in layer l , with hidden state $H^{(l-1)}$, the attention pattern is

$$A_h^{(l)} = \text{softmax} \left(\frac{H^{(l-1)} W_h^Q (W_h^K)^T (H^{(l-1)})^T}{\sqrt{d_k}} \right), \quad (2)$$

and the head output is $A_h^{(l)} H^{(l-1)} W_h^V W_h^O$. Following transformer-circuit notation [5], the QK circuit $W_h^Q (W_h^K)^T$ controls attention scores, while the OV circuit $W_h^V W_h^O$ controls what is written to the residual stream. For nucleotide models, tokenization matters: a motif position may correspond to one nucleotide token, a fixed k -mer token, or a learned BPE span.

Strict motif-detector criterion. An attention head is treated as a candidate biological motif detector only when three signals converge. It must assign elevated attention mass to motif-support positions, its QK scores must rank motif-support tokens above matched background tokens, and patching its activation must restore the task-relevant scalar in motif-destroying counterfactuals. The registered thresholds used here are in Table 1. These thresholds intentionally make attention alone insufficient.

Activation patching. For a clean sequence x_c and motif-destroyed corrupted sequence x_{corr} , we patch a head output from the clean run into the corrupted run. The restoration metric is

$$\text{PM}(h, l) = \frac{f(x_{\text{patched}}) - f(x_{\text{corr}})}{f(x_c) - f(x_{\text{corr}})}, \quad (3)$$

Table 1: Registered evidence criteria for candidate motif circuits.

Claim	Statistic	Threshold
Motif localization	Attention enrichment ratio ρ_h	$\rho_h \geq 2.0$
Motif ranking	QK/motif Pearson correlation	$r \geq 0.5, p < 0.05$
Causal importance	Patching metric $PM(h, l)$	$PM \geq 0.5$
Strong causal importance	Same	$PM \geq 0.75$
Residual decodability	Probe AUROC or macro- F_1	≥ 0.80 or $+0.10$
Robustness	Matched negative controls	No matched artifact

where f is a scalar target such as a probe decision function or class logit. $PM = 0$ indicates no recovery, $PM = 1$ indicates full recovery of the clean effect, and values above one indicate over-restoration rather than simple full recovery [6].

Residual probes and controls. Linear probes are trained on frozen mean-pooled residual vectors. Their role is diagnostic: a high probe score establishes linear decodability, not that the probe direction is causal or used by the model. We therefore compare residual probes with GC-content-only classifiers, position-only classifiers where coordinates are available, GC-matched test subsets, shuffled-label probes, and GC-content distribution shifts. This framing follows linear-representation work while preserving the weaker interpretation that probes support [11].

4 Experimental Setup

The primary model is zhihan1996/DNABERT-2-117M, an encoder-style genomic transformer with learned BPE tokenization. The comparison backend is InstaDeepAI/nucleotide-transformer-v2-100m-multi-species, used as a tested fixed-6mer Nucleotide Transformer checkpoint [4]. The run exports DNABERT-2 QK and OV matrices for all 12 layers and 12 heads per layer, caches residual vectors for layers 0, 5, and 11, and uses custom Hugging Face forward hooks for attention-output patching because the installed TransformerLens path does not expose all required DNABERT-2 per-head tensors.

The residual-probe tasks are TATA promoter, non-TATA promoter, splice donor, and splice acceptor classification. The configured splits contain 5,062/212 train/test examples for TATA promoters, 30,000/1,372 for non-TATA promoters, and 30,000/3,000 for each splice task. The strict CTCF scan uses 51,249 prepared GM12878 CTCF peak sequences and the JASPAR MA0139.1 CTCF motif, aligned to exact overlapping DNABERT-2 BPE token spans. Batch patching uses motif-destroying clean/corrupted pairs for promoter-TATA and splice-donor tasks.

The current artifact does not include a fine-tuned DNABERT-2 sequence-classification head. To avoid treating probes as end-to-end model performance, we therefore report two raw-sequence predictive baselines before the mechanistic evidence: a GC-content logistic classifier using GC fraction, GC skew, and sequence length, and a TF-IDF 3–6-mer logistic classifier. Frozen DNABERT-2 residual readouts are shown beside these baselines only as representation context.

The pipeline writes small review-facing CSV, JSON, and figure artifacts while treating large activation, circuit-matrix, token-score, and sparse-autoencoder files as reproducible generated outputs. The full run takes about eight hours on the tested CUDA setup. The most expensive stages are the cross-model comparison and strict all-layer CTCF scans, so the appendix includes both full reproduction commands and a reduced smoke-run command for checking the code path on a smaller subset.

5 Results

Predictive performance context before mechanism. Table 2 separates task predictability from mechanistic evidence. The raw-sequence baselines are strong, especially on promoter tasks: a simple GC-content classifier already reaches 0.8955–0.9088 AUROC, and a 3–6-mer classifier reaches 0.9297–0.9406 AUROC. Splice tasks are less explained by composition alone, but k-mer baselines still recover substantial signal. The frozen DNABERT-2 layer-11 readout is included to show that the pretrained representation is predictive, but it remains a residual-decoding result, not a fine-tuned classifier-head result and not a circuit explanation.

Table 2: Predictive task context before mechanistic analysis. GC-only and 3–6-mer rows are raw-sequence baselines. Frozen DNABERT-2 readout is a layer-11 residual classifier and is reported as diagnostic decodability evidence, not as end-to-end task performance.

Task	Train	Test	GC AUROC	GC AUPRC	GC Acc.	k-mer AUROC	k-mer AUPRC	k-mer Acc.	DNABERT readout AUROC
promoter_tata	5,062	212	0.8955	0.9040	0.8019	0.9297	0.9378	0.8255	0.9137
promoter_no_tata	30,000	1,372	0.9088	0.9218	0.8316	0.9406	0.9482	0.8652	0.9383
splice_sites_donors	30,000	3,000	0.6560	0.6892	0.6013	0.8185	0.8093	0.7430	0.8954
splice_sites_acceptors	30,000	3,000	0.6361	0.6343	0.5883	0.7956	0.7792	0.7220	0.8847

Table 3: DNABERT-2 layer-11 probe-control summary. Numeric columns report AUROC; random-label values are means over eight shuffled-label runs.

Task	Resid.	GC-only	Pos.-only	GC-match	Rand.	GC-shift range
promoter_tata	0.9137	0.8956	0.4136	0.9137	0.4700	0.6182–0.8030
promoter_no_tata	0.9383	0.9088	0.3865	0.9383	0.5129	0.6312–0.8368
splice_sites_donors	0.8954	0.6560	0.4414	0.8944	0.5064	0.8432–0.8669
splice_sites_acceptors	0.8847	0.6361	0.4461	0.8838	0.5005	0.8573–0.8580

Residual decodability is strong but not uniformly clean. DNABERT-2 layer-11 residual probes exceed the registered residual-decodability threshold for all four tasks. Table 3 combines headline AUROC values with the most important controls. Random-label probes collapse to chance, and position-only metadata does not explain the results. The splice-site tasks survive stronger checks: donor and acceptor residual probes remain about 0.25 AUROC above GC-only baselines on matched subsets and remain strong under GC shifts. The promoter tasks require caution because GC-only baselines are already high; their residual-probe advantage over GC-only controls is small on the matched comparison.

This distinction changes the interpretation of the headline probe table. The splice-site result is evidence that DNABERT-2’s frozen residual stream contains linearly accessible information beyond simple GC composition or coordinate metadata. The promoter result is weaker mechanistically: it remains a valid decodability result, but the controls show that promoter-associated composition can explain much of the signal. This is why the paper avoids claiming that the promoter probes have isolated a clean causal promoter motif direction.

The strict CTCF motif-detector test fails. The CTCF scan produced 281,915 motif-support tokens and 1,843,874 finite token positions after BPE span correction. Across all 144 DNABERT-2 heads, the best QK-to-motif correlation is $r = 0.3004$, below the registered $r \geq 0.5$ criterion. The best matched attention-enrichment ratio is $\rho_h = 1.3130$, below the registered $\rho_h \geq 2.0$ criterion. The tested Nucleotide Transformer backend also fails to recover a strict CTCF candidate, with best $r = 0.0192$ and best $\rho_h = 1.00009$. Thus, the tested fixed-6mer backend does not support the CTCF motif-local head hypothesis under the same criteria.

The negative CTCF result is not a failure of the pipeline; it is the point of using a registered evidence stack. With more than 1.8 million finite token positions, small correlations can be statistically nonzero while remaining too small to support a mechanistic claim. The all-layer enrichment scan similarly prevents a common attention-overinterpretation mistake: a head can have visible structure without assigning enough mass to matched motif-support positions to count as a motif-local detector.

Auxiliary promoter/splice patching gives task-specific causal signals. Batch promoter-TATA patching over 327 usable pairs identifies layer 4 head 8 with mean PM = 1.4029; layer 6 head 8 and layer 2 head 7 are also high. Because PM > 1 overshoots the clean-minus-corrupted effect, this is a methodologically sensitive causal signal rather than a simple full-recovery story. Batch splice-donor patching over 500 pairs identifies layer 1 head 8 with mean PM = 0.5485, a weaker but threshold-crossing effect. These interventions support task-relevant internal computation, but they do not identify a strict CTCF motif detector because the CTCF QK and enrichment screens have no passing head.

OV, SAE, and cross-model audits remain conservative. The TATA OV readout audit for layer 2 head 7 finds only weak alignment with the trained TATA probe direction: the top output-write singular-vector cosine is 0.1261 and the probe self-gain through the OV matrix is -0.0326 . Sparse-autoencoder

Table 4: Target-alignment ledger. CTCF is the primary strict motif-detector chain. Promoter and splice results are auxiliary evidence for task-specific residual decodability and causal signal, not substitutes for a CTCF proof.

Target	Performance	Probe	QK	Enrichment	Patching	OV	SAE	Role
CTCF	51,249-sequence strict scan; no classifier target	No CTCF residual classifier used as strict evidence	Best $r = 0.3004$; fail	Best $\rho_h = 1.3130$; fail	No candidate promoted after failed QK/enrichment	No aligned OV claim	Best cosine 0.1158; weak	Primary strict chain
Promoter-TATA	GC 0.8955; k-mer 0.9297; readout 0.9137 AUROC	Decodable, but close to strong baselines	Not aligned current fact	Not target-aligned in current fact	Best PM = 1.4029; over-restores	L2H7 out-put cosine 0.1261; weak	Not promoter-specific	Auxiliary causal evidence
Splice donor	GC 0.6560; k-mer 0.8185; readout 0.8954 AUROC	Decodable beyond GC k-mer	be-Not aligned current fact	Not target-aligned in current fact	Best PM = 0.5485; partial	Not audited	Not splice-specific	Auxiliary causal evidence

Table 5: Strict mechanistic evidence summary from the completed run.

Evidence channel	Executed test	Best observed result	Met?
Residual decodability	Layer-11 probes on four benchmark tasks	AUROC range 0.8847–0.9383	Yes
Probe controls	GC, position, matched, random-label, and shift controls	Splice robust; promoter close to strong GC baselines	Mixed
CTCF QK alignment	144 DNABERT-2 heads vs. BPE-correct CTCF scores	Best $r = 0.3004$	No
CTCF enrichment	Matched motif/background attention mass	Best $\rho_h = 1.3130$	No
Promoter-TATA patching	327 batch denoising pairs	Best mean PM = 1.4029	Yes
Splice-donor patching	500 batch denoising pairs	Best mean PM = 0.5485	Partial
OV readout audit	TATA probe direction vs. OV singular vectors	Top cosine 0.1261; self-gain -0.0326	Weak
CTCF SAE search	Residual and MLP sparse features	Best motif cosine 0.1158	Weak
Strict CTCF detector	QK, enrichment, and causal evidence in one head	No passing QK/enrichment candidate	No

CTCF feature search is also weak, with best residual and MLP motif cosines of 0.0884 and 0.1158. In the protocol-level cross-model comparison, DNABERT-2 residual probes outperform the tested Nucleotide Transformer backend on the same tasks, but this is a checkpoint- and protocol-specific result, not a general claim about tokenization families or all Nucleotide Transformer variants.

6 Limitations

This study is intentionally scoped to small and medium genomic transformers and does not claim exhaustive explanation. Current large-model circuit-tracing work also emphasizes partial recovery of computation rather than complete mechanistic accounts [2, 9]. The strict CTCF failure applies to the selected models, CTCF intervals, motif scorer, BPE span alignment, and registered thresholds; it is not an impossibility theorem for all models or motifs.

Linear probes remain limited evidence. They can reveal decodable information while exploiting correlated sequence composition, split artifacts, or metadata. The added GC, position, matched, random-label, and shift controls reduce this risk, especially for splice-site tasks, but promoter decodability remains entangled with GC composition. Activation patching is causal but incomplete: head-level restoration does not characterize the full circuit, and over-restoration can reflect denominator sensitivity, probe geometry, or out-of-distribution patched states. Finally, biological annotations are incomplete, and a single JASPAR PWM cannot represent all relevant regulatory mechanisms.

7 Conclusion

MINTS turns a common interpretability temptation into a falsifiable test: before calling an attention head a biological motif detector, require aligned QK scores, motif-local attention enrichment, and causal restoration. On DNABERT-2, promoter and splice-site labels are strongly linearly decodable from late residual states, and splice-site decodability survives simple composition and position controls. But the strict CTCF motif-detector proof fails across all tested heads, and the task-specific

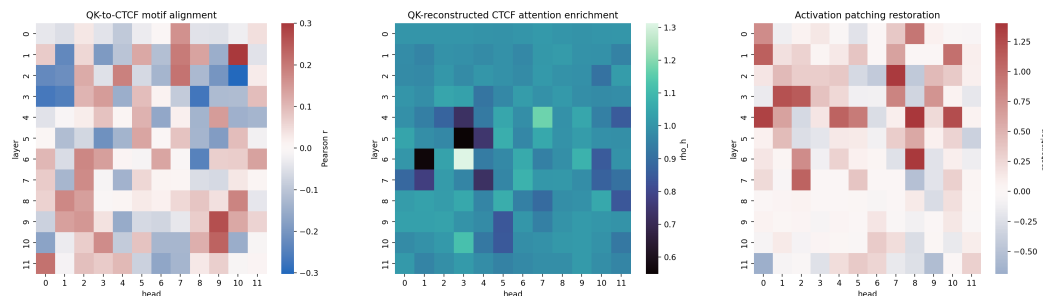


Figure 1: Mechanistic evidence heatmaps. Left: DNABERT-2 CTCF QK-to-motif correlations. Middle: matched attention-enrichment ratios. Right: promoter-TATA batch patching. The CTCF heatmaps show a negative strict-proof result, while the TATA patching heatmap shows task-specific causal over-restoration.

patching effects do not change that conclusion. The contribution is therefore both positive and negative: a reproducible evidence standard for nucleotide-transformer interpretability, and a concrete demonstration that strong representation-level decodability should not be upgraded into a circuit-level biological claim without stronger evidence.

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237 A Theoretical Details

238 **Definition 1** (Biological Motif Detector). *An attention head h in layer l functions as a biological*
 239 *motif detector for motif m if there exists a nonempty set of relevant query positions $\mathcal{Q}_m$ and a*
 240 *threshold τ such that, for each $i \in \mathcal{Q}_m$,*

$$\mathbb{E} \left[\sum_{j \in \text{supp}_m(x)} A_h^{(l)}[i, j] \mid \text{supp}_m(x) \neq \emptyset \right] \geq \tau \quad (4)$$

241 *and the corresponding expectation conditioned on $\text{supp}_m(x) = \emptyset$ is below τ . Here $\text{supp}_m(x)$ is the*
 242 *set of key positions whose local sequence windows instantiate motif m .*

243 **Theorem 1** (Layer-1 Motif Detection via QK Circuits). *Consider a layer-1 attention head operating*
 244 *on $H^{(0)}$. Let $s_m(j)$ denote a scalar motif score associated with key position j . If for a fixed query*
 245 *position i there are constants $a_i \in \mathbb{R}$ and $\gamma > 0$ such that*

$$q_i^T k_j = a_i + \gamma s_m(j) \quad (5)$$

246 *for all admissible key positions j , then the induced attention distribution is monotonically increasing*
 247 *in $s_m(j)$. If motif-bearing positions are separated from background positions by a positive margin in*
 248 *s_m , the head is a motif detector for m .*

249 *Proof.* The attention logit is $\ell_{ij} = q_i^T k_j / \sqrt{d_k} = a_i / \sqrt{d_k} + (\gamma / \sqrt{d_k}) s_m(j)$. Since $\gamma / \sqrt{d_k} > 0$,
 250 scaling preserves the ordering of motif scores. The exponential and softmax are strictly monotone in
 251 each logit when other logits are fixed, so positions with larger motif scores receive larger attention
 252 weights. A positive separation margin therefore induces an attention-mass threshold that separates
 253 motif-bearing from background positions. $\square$

254 **Lemma 1** (Approximate Extension Under Linear Decodability). *Suppose that at layer $l - 1$ there*
 255 *exists a vector u_m such that*

$$u_m^T h_j^{(l-1)} = s_m(j) + \epsilon_j, \quad |\epsilon_j| \leq \delta. \quad (6)$$

256 *If a head's QK score is affine in this decodable direction and the clean motif/background score margin*
 257 *is greater than 2δ , then the attention ordering between motif and background positions is preserved.*

258 *Proof.* For motif-bearing j_m and background j_b , the score difference is proportional to $s_m(j_m) -$
 259 $s_m(j_b) + \epsilon_{j_m} - \epsilon_{j_b}$. The perturbation term lies in $[-2\delta, 2\delta]$. A clean separation greater than 2δ keeps
 260 the sign of the score difference fixed, so softmax preserves the motif/background ordering. $\square$

261 B Implementation and Reproducibility Details

262 The completed run started at 13:27:07 local time on April 14, 2026 and ended at 21:26:30 local time
 263 the same day. The manifest reports 28,760.213 seconds, or 7.989 hours. The longest steps were
 264 cross-model tokenization comparison, strict mechanistic proofs, and systematic causal intervention.
 265 The inspected result tree contains 125 files totaling about 4.71 GB, including CSV/JSON summaries,
 266 figures, NPZ archives, and sparse-autoencoder checkpoints.

267 The anonymized project repository is available at [https://anonymous.4open.science/r/](https://anonymous.4open.science/r/MINTS/)
 268 MINTS/. It contains the project code, configuration, local paper source, and reproduction instructions.
 269 The repository currently serves as the single source of truth for the submitted code; an explicit
 270 repository license has not yet been added.

271 The main reproduction commands are:

```
272 git clone https://anonymous.4open.science/r/MINTS/ MINTS
273 cd MINTS
274 python -m pip install -r requirements.txt
275 python main.py
276 python main.py --only-probe-controls
```

277 A smaller smoke run is:

```

278 python main.py --max-probe-train 512 --max-probe-test 256
279     --max-qk-alignment-sequences 128
280     --max-cross-model-qk-alignment-sequences 128
281     --max-feature-search-sequences 128 --sae-epochs 1

```

282 Important verification artifacts include:

- 283 • results/pipeline_run.json
- 284 • results/tables/linear_probe_metrics.csv
- 285 • results/tables/linear_probe_controls.csv
- 286 • results/tables/downstream_task_performance.csv
- 287 • results/tables/target_alignment_table.csv
- 288 • results/tables/review_issue_matrix.csv
- 289 • results/qk_alignment/ctcf_qk_alignment.csv
- 290 • results/enrichment/ctcf_qk_alignment_matched_attention_enrichment.
- 291 csv
- 292 • results/patching/promoter_tata_batch_dnabert_activation_patching.csv
- 293 • results/patching/splice_sites_donors_batch_dnabert_activation_
- 294 patching.csv

295 The heatmaps used in the paper are generated under results/figures/.

296 C Additional Results

297 The full configured-split residual probe AUROCs are 0.9137 for TATA promoters, 0.9383 for non-
298 TATA promoters, 0.8954 for splice donors, and 0.8847 for splice acceptors. The corresponding
299 AUPRC values are 0.9241, 0.9475, 0.9049, and 0.8954.

300 D Review-Triage Matrix

301 Table 6 records the review issue matrix used to guide the revision. The anonymized reviewer IDs are
302 treated as issue sources rather than as claims about reviewer intent.

Table 6: Review issue matrix.

ID	Issue	Concern	Resolution	Status
mV6D	Clarity	Paper read as several analyses.	Abstract/introduction now state one calibrated evidence-standard claim; Results opens with task context and a target ledger.	Addressed
mV6D	Target mismatch	Promoter/splice evidence blurred into CTCF claims.	Promoter/splice patching is auxiliary; CTCF remains the strict chain.	Addressed
C5jL	Missing performance	Mechanistic claims lacked predictive context.	Added GC and 3–6-mer baselines plus frozen-readout context before mechanism.	Partial
C5jL	MI contribution	Venue contribution needed sharpening.	Contribution is now a falsifiable standard for motif-detector claims.	Addressed
C5jL	Thresholds	Registered thresholds need scale/sensitivity.	Kept as conservative standards here; sensitivity/null analysis remains future work.	Deferred
ndFG	Undefined biology	Motifs and biological terms need more support.	Motifs are motivated in the introduction; a full glossary and token-mapping description remain future work.	Partial
ndFG	Limited scope	Conclusions could overstate selected models/motifs.	Limitations and target ledger state the scoped negative CTCF result.	Addressed

303 For promoter-TATA batch patching, 327 token-shape-preserving pairs were usable with no near-zero
304 denominator failures. The best mean score was layer 4 head 8 with $PM = 1.4029$, followed by layer
305 6 head 8 with $PM = 1.3791$ and layer 2 head 7 with $PM = 1.3642$. For splice donors, 500 pairs
306 were usable with no denominator failures, and the strongest mean restoration was layer 1 head 8 with
307 $PM = 0.5485$.

308 For the tested Nucleotide Transformer backend, residual-probe AUROCs were 0.6502, 0.6976,
309 0.5695, and 0.5647 on the same four tasks. DNABERT-2 therefore outperformed this backend by

0.2408–0.3259 AUROC in this protocol. This comparison should not be read as a general ranking of tokenizers or genomic transformer families.

E Declaration of LLM Usage

LLM-assisted tools were used during the project for mathematical exposition refinement, debugging support, and code-generation drafts through Codex with state-of-the-art GPT-5.4/GPT-5.5-class models. The research direction, core problem decomposition, step-by-step implementation decisions, and final scientific interpretation were directed by the author. The author reviewed the generated code, checked it for errors and failure modes, corrected mistakes, and takes responsibility for the final manuscript, code, and claims.

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